

State Machine Question (100 marks)

Given the problem description below, specify the **state machine** for the Thermomix® TM6 kitchen appliance. **Show the states as well as transitions including events, guards, and actions.** You do **not** need to specify parameters for events, guards, and actions. If you use private methods for guards or actions in your state machine, use **meaningful** names for those methods. You do **not** need to define the method bodies of those methods.

Problem Description: Thermomix® TM6

The Thermomix® TM6 is an all-in-one kitchen appliance that preps ingredients and cooks them to perfection.

On delivery, the Thermomix® TM6 is set to transportation mode. When the selector (button) is pressed to start up the Thermomix® TM6 for the first time, the transportation mode is automatically deactivated, and the home screen is shown. To turn the Thermomix® TM6 off, hold the selector down for at least five seconds until a message appears to confirm that the Thermomix® TM6 is switching off. You can then release the selector. If the Thermomix® TM6 has been turned off, pressing the selector turns it back on and the home screen is shown. To save energy, the Thermomix® TM6 switches off automatically after 15 minutes when not in use. A message appears for the last 30 seconds, allowing automatic shutdown to be canceled and the home screen to be shown (by selecting cancel on the appliance's screen or by removing the cooking bowl).



To cook a meal, select a recipe on the screen and then select start to follow the step-by-step instructions. First, add ingredients as instructed. The integrated scale weighs them and allows the next step only if the correct amount has been added. Continue to the next step by selecting next on the screen. The Thermomix® TM6 chops the ingredients for as long as and at the speed required for the recipe. When the chopping step is done, select next for the Thermomix® TM6 to start the cooking step. Again, the Thermomix® TM6 cooks the meal at the temperature and time required for the recipe. At the end of any recipe step, the Thermomix® TM6 may prompt you to add further ingredients, which are then again weighed, chopped, and cooked. After the last step, the Thermomix® TM6 informs you that the meal is ready to be served. When the cooking bowl is removed, the Thermomix® TM6 returns to the home screen. It is not possible to cook a meal if the cooking bowl is not correctly placed on the Thermomix® TM6.

The selector button raises the following events: **selectorPressed** (the selector is pressed and released immediately), **selectorHeld** (the selector is pressed and not released immediately), and **selectorReleased** (the selector is released after being held).